

Nathan Tang

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Canadian citizen eligible for U.S. employment with TN status, open to relocation.

EXPERIENCE

People Inc. (Dotdash Meredith) *Backend Software Engineer II* May 2022 – Present

- One of three founding engineers building a global configuration and telemetry platform (Node.js, Java, SQL, Kafka, Vue.js), defining core architecture and distributed service design, generating **\$1M/day in revenue**.
- Led backend architecture and service design, scaling APIs to handle **millions of requests daily** through schema modeling, query optimization, caching strategy, and authentication systems.
- Built a distributed event-driven pipeline using Kafka to ingest **7M+ daily events**, enabling analytics in BigQuery, supporting fault tolerance, backpressure handling, and large scale data analysis.
- Architected a data processing pipeline for dynamic bundling and recursive configuration resolution, **reducing payload size by 30%** and improving response latency across clients.
- Defined a deduplication and configuration tiering framework, consolidating overlapping rules to **reduce database footprint by 40%**, lowering compute and memory overhead.
- Designed scalable filtering and ranking algorithms to serve contextually relevant configurations per request, maintaining **linear time complexity under large datasets** and high request volume.
- Established tooling to automate API schema-to-DTO generation, creating type-safe backend-frontend interfaces, accelerating development, and reducing runtime errors through consistent contracts.
- Built reusable components adopted **across 40+ brands**, enabling flexible configuration at scale and improving viewability while **sustaining 95+ on Core Web Vitals** scores.

Aquatic Biosphere Society of Canada *Full Stack Engineer (Contract)* January 2024 – April 2024

- Led end-to-end architecture of core APIs (Python, PostgreSQL), defining data models and service boundaries for user and CMS functionality, integrating Google OAuth across game and admin interfaces.
- Architected and optimized an OpenCV-based image processing pipeline, **reducing latency from ~10s to <1s** by redesigning data flow and tuning for real-world bandwidth and device constraints.
- Owned cloud deployment strategy on Azure, provisioning web and database services to support live gameplay traffic and compute-intensive image workloads with production reliability.

Snap & Tow *Backend Engineering Intern* March 2022 – May 2022

- Built the backend from scratch for an early-stage marketplace, implementing real-time geospatial matching (Google Maps API) to power proximity-based dispatch, forming the core service logic of the platform.

PROJECTS

thechoice.fm | PHP, Node.js, Garage (S3)

- Architecting an on-demand radio service, enabling synchronized playback via a distributed playlist system.
- Established a timed-based scheduling service that evaluates a hierarchical rule tree to populate future broadcast slots, resolving user-selected slot overrides while preserving scheduling constraints.

Homelab | Linux, Proxmox, Docker

- Architected and deployed a self-hosted infrastructure platform using Proxmox and Docker, orchestrating VMs and containers to support media services, networking, and development workloads.
- Designed secure global access architecture, enabling reliable remote connectivity and service exposure.

EDUCATION

University of Alberta | Bachelor of Science in Computing Science

TECHNICAL SKILLS

Languages	Go (Golang), Java, TypeScript, Python, SQL, JavaScript, C, HTML/CSS
Databases	PostgreSQL, Redis, MySQL, MongoDB, Firebase
Technologies	Kafka, Docker, AWS, Azure, Node.js, Express.js, Django, React/Vue/Svelte, Git